

Root cause analysis (RCA) across industries: Common principles, traditional methods, commercial offerings landscape and case studies

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RCA is mature in industries with one or more of three characteristics: *high cost of failure, strong safety/regulatory requirements, or complex systems where failures can have many interacting causes.* Example industries include manufacturing, automotive, aerospace & defense, healthcare, pharmaceuticals & life sciences, information technology, construction, energy & utilities, oil & gas, chemical, mining, transportation & railways, logistics & supply chain, telecommunications, banking & financial services, retail & e-commerce, food & beverage, electronics & semiconductors.

Section 1. Five Core Principles of Root cause analysis

Although RCA methodologies vary by industry, there are [five foundational principles](#) that distinguish root cause analysis from solely identifying symptoms or assigning responsibility.

- 1. **Causality (Identify the underlying cause, not just the symptom):** RCA seeks to understand the cause and effect relationships that produced an undesirable outcome. Complex problems may have multiple causal and contributing factors rather than a single root cause.
- 2. **Specificity (Identify causes that are actionable):** A useful root cause must be sufficiently specific to support corrective action. Broad explanations such as human error, poor communication, or process failure generally require further investigation. Effective RCA continues until the analysis identifies underlying causes that can reasonably be addressed through changes to processes, controls, resources, technology, training, or other operating conditions.
- 3. **Evidence (Establish causes through data):** RCA conclusions should be supported by factual evidence rather than assumptions or intuition. Relevant evidence may include operational data, project records, timelines, interviews, observations, system logs, process measurements, and other documentation. Potential causes should be evaluated and, where possible, verified before corrective actions are selected.
- 4. **Prevention (Focus on preventing recurrence):** The purpose of RCA is not merely to explain what happened, but to use that understanding to prevent the problem from happening again. Identified root causes should lead to corrective or preventive actions, followed by verification that those actions have reduced or eliminated the underlying source of the problem.
- 5. **Systems Thinking (Examine the conditions that allowed the problem to occur):** RCA looks beyond individual actions to understand the broader system in which an event occurred. Root causes may involve processes, procedures, equipment, information flows, organizational structures, controls, resources, or interactions among multiple factors. A systems perspective helps identify vulnerabilities that can be addressed through sustainable improvements rather than focusing solely on individual error.

Section 2. Engineering and construction RCA business use cases by insurance-claim frequency and severity

The following table compiles share by claim count and value from **13,599 [engineering and construction insurance claims processed by Allianz from July 2013 through July 2018](#)**, valued at approximately €7.9 billion. It is not a universal construction-industry dataset, however, can be useful for understanding the types of use cases for RCA encountered.

Ranking by insurance-claim frequency and severity

Combined priority	RCA cause category	Share by claim count	Share by claim value	After-effects of the cause

1	Defective products and materials	27%	17%	High frequency, major rework, latent defects, failed components, and supplier/subcontractor responsibility
2	Storm and natural-catastrophe damage	10%	10%	Weather can damage large exposed areas and may trigger major delay, mitigation, and coverage questions
3	Faulty workmanship and maintenance	9%	15%	Often causes recurring defects, water intrusion, structural issues, and disputes over responsibility
4	Water damage and leakage	8%	4%	Very common, can spread through concealed areas, and requires difficult source-and-timing analysis
5	Fire and explosion	7%	27%	Less frequent than defects or storms, but the most severe category by total claim value
6	Design and engineering errors	Included within broader defect categories	Included within broader defect categories	Can cause rework, delay, structural problems, code issues, and professional-liability claims
7	Equipment and machinery failures	Not separately quantified in the cited Allianz table	Not separately quantified	Can cause property damage, injury, interruption, and questions about maintenance or operator error
8	Theft, vandalism, and site-security failures	Not separately quantified	Not separately quantified	Common during partially secured construction and may involve inventory, access, and security-control failures

9	Construction accidents and unsafe conditions	Often handled under workers' compensation or general liability rather than engineering property claims	Not directly comparable	High injury, fatality, liability, and regulatory exposure
10	Schedule, coordination, and change-management failures	Usually embedded in delay, professional liability, or builder's-risk claims	Often difficult to isolate	Can create delay, acceleration cost, concurrent-delay disputes, and business-interruption losses

Section 3. Overview of traditional RCA methods

These traditional RCA methods below are covered as part of RCA certifications and can be applied to both manual debugging and AI-assisted analysis. They provide a structured approach to analyzing collected evidence, identifying contributing factors, and determining the underlying root cause.

Tool/Technology	Description
5 Whys	Identifies the root cause by asking “Why” multiple times
Fishbone/Ishikawa diagram	The fishbone diagram/Ishikawa diagram, is a root cause analysis tool that helps teams identify possible causes of a problem across six categories: methods, people, material, machine, measurement, and environment .
Fault tree analysis	Identifies the specific combination of events that caused a problem by tracing contributors through logical relationships until the root causes are reached. Probabilities are then calculated through the tree to determine the likelihood of the problem occurring.
Pareto analysis	The Pareto chart applies the 80/20 rule, which suggests that roughly 80% of problems come from 20% of causes. This helps identify and prioritize the issues with the greatest impact.

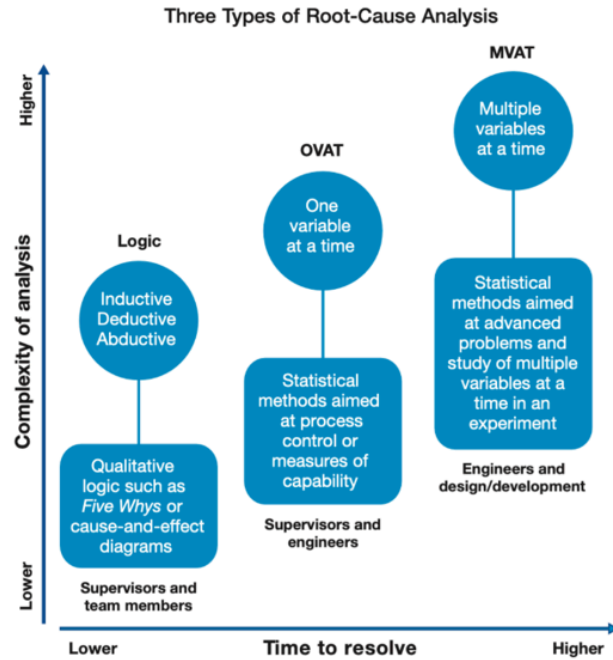


Fig. 1 The image has been taken from Lean.org

3.1 Five Whys

The 5 Whys is a structured method for identifying the underlying cause of a problem by repeatedly asking “Why?” instead of stopping at the immediate symptom. The process begins by assembling people familiar with the issue and clearly defining the problem. The team then asks why the problem occurred and continues questioning each preceding cause, typically five times or until a root cause is identified. Answers should be supported by facts and data rather than assumptions. Once the root cause is established, the team defines corrective actions, assigns ownership, monitors whether those actions resolve the issue, and documents the analysis for future learning. **Though the name suggests five times, however investigators should continue beyond that till they reach the initial cause in the chain.**

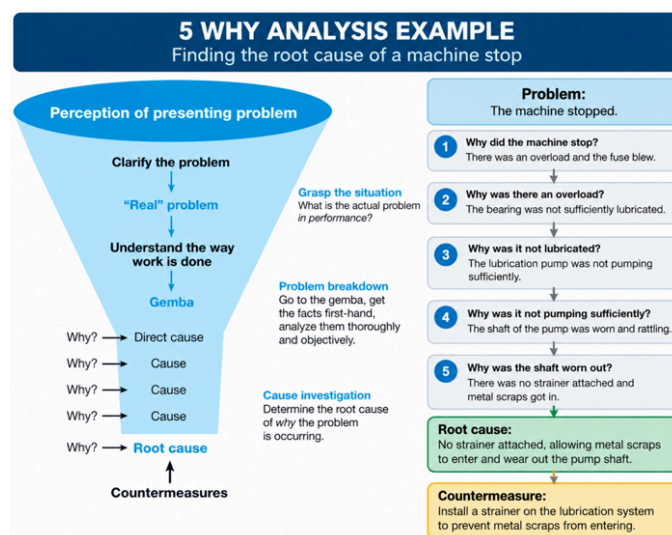


Fig. 2 The image has been taken from Lean.org

3.2 Ishikawa/ Fishbone diagram

An Ishikawa/ Fishbone or cause and effect diagram, helps teams systematically identify and explore the potential causes of a problem. The effect is placed at the head of the diagram, while possible causes branch out into categories such as people, methods, machines, materials, measurement, and environment. Teams then break these categories into more specific contributing factors to uncover underlying root causes. By visually organizing relationships between a problem and its potential causes, the fishbone diagram encourages collaborative brainstorming, brings together different perspectives, and helps teams move beyond symptoms to identify areas where corrective actions can have the greatest impact.

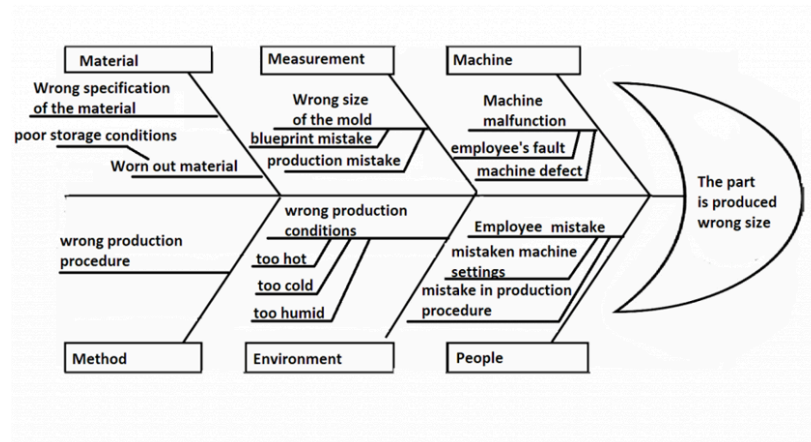


Fig. 3 Example of applying the Ishikawa/ Fishbone diagram for investigating the root cause of a defect. It is taken from [lean six sigma definition](#).

Industry Examples:

- Automotive manufacturing (Ford, identifying transmission system inefficiencies; Toyota, identifying root causes of production defects)
- Food and beverage (Starbucks, streamlining production and identifying supply chain delays) aerospace (NASA, investigating technical anomalies in missions), aviation (Delta Air Lines, proactively preventing flight delays)
- Technology (Microsoft, identifying process inefficiencies; tech startups, improving software debugging and reducing code errors) and
- Electronics manufacturing (Samsung, improving product design, innovation, and quality).

3.3 Fault Tree Analysis(FTA) and Failure Modes and Effects Analysis (FMEA)

Fault Tree Analysis (FTA) is a top down, **deductive root cause analysis** method used to understand how an undesired event or system failure could occur. The analysis begins with a defined top event, such as poor quality concrete pouring, and works downward by identifying the events and conditions that could cause it. These causes are progressively decomposed until the analysis reaches basic or root causes. Logic gates specify how combinations of causes lead to higher level events. The resulting tree visually maps potential failure pathways, helping teams identify critical failure points, prioritize corrective actions, and determine where controls could prevent the failure from occurring.

An OR gate means any one of the connected causes is sufficient to produce the event above it. For example, if poor concrete quality can result from either an inadequate mix design, improper placement, or environmental factors independently, these branches would be connected through an OR gate. An AND gate means all connected causes must occur together for the event above them to occur. If a failure only happens when both inadequate equipment and incorrect operating conditions are present, the two causes would be connected with an AND gate. When exactly one of the connected events must occur, an exclusive OR or XOR gate is applied.

Primary causes are the first level causes directly beneath the top event. In the poor quality concrete example below taken from Mitti, they include insufficient concrete mix design, improper concrete placement and consolidation, and environmental factors. *Secondary causes*

explain why the primary causes occurred and take the analysis one level deeper. *Tertiary causes* go another level deeper toward fundamental or actionable root causes. In the construction example, these include lack of quality assurance processes, poor communication between engineers and suppliers, inadequate supervision, failure to follow industry standards, and failure to implement measures for adverse weather. The decomposition can continue beyond three levels until meaningful basic or root causes are reached.

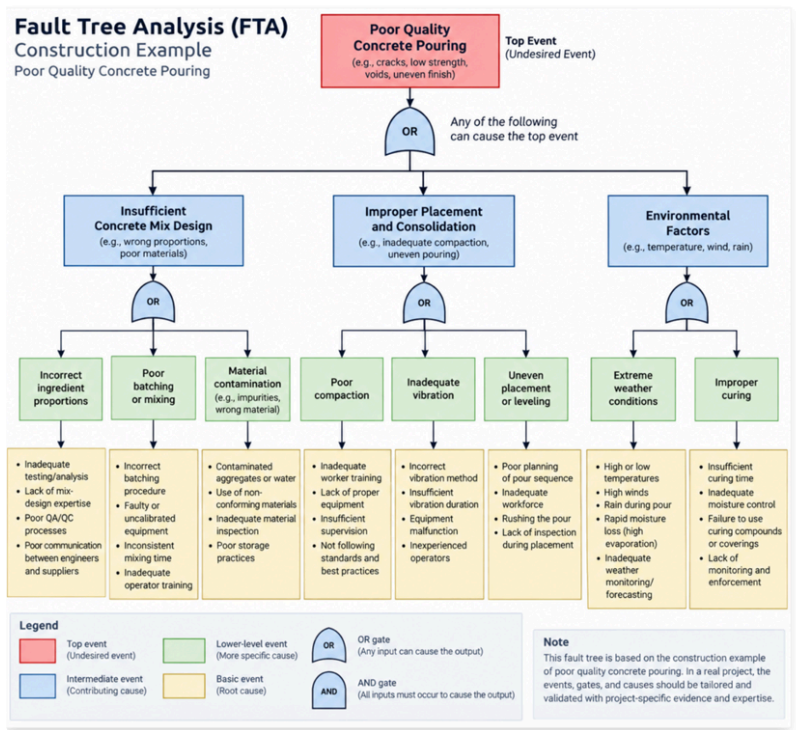


Fig. 4 Example of applying Fault Tree Analysis for a construction use case. The case is described in this [Mitti example scenario](#).

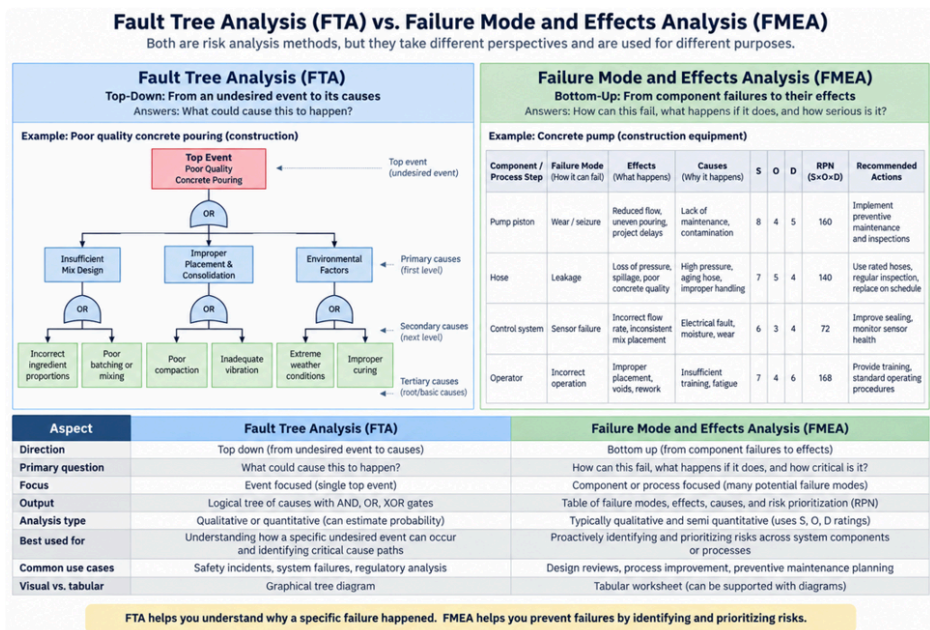


Fig. 5 Comparing FTA vs FMEA

3.4 Pareto Analysis

Pareto charts help teams identify and prioritize the **small number of causes responsible for the majority of problems**, based on the Pareto or 80/20 principle. First, define the problem and collect data on the frequency of potential causes over a specified period. Next, tally occurrences, sort the causes from most to least frequent, and display them as bars. Add a cumulative percentage line to show which causes collectively account for at least 80% of the occurrences. Teams can then focus RCA efforts on the highest contributing categories, investigate those categories in greater detail, identify their underlying causes, and prioritize corrective actions where they can have the greatest impact.

Section 4. Commercial products for RCA

4.1 TapRooT®

- **Root cause analysis problem:** TapRooT addresses the challenge of moving beyond the immediate cause of a construction or industrial incident to identify the underlying organizational, human performance, equipment, procedural, and management system factors that allowed it to occur. It is primarily used for safety incidents, near misses, equipment failures, quality events, and proactive risk investigations.
- **Data processed:** TapRooT is primarily investigator and evidence driven. It processes incident facts, observations, interviews, documents, equipment information, timelines, and other evidence collected during an investigation. Its structured Root Cause Tree® and associated tools help investigators classify causal factors, determine root causes, develop corrective actions, and identify recurring patterns across investigations.
- **Client coverage:** TapRooT serves construction, manufacturing, energy, utilities, nuclear, mining, aviation, and other high-risk industries. Publicly referenced users include Intel, Arkema, Oceaneering, and Air Canada Maintenance, with customers across North America, Europe, Asia, and Australia.

4.2 SmartPM & Oracle P6

Oracle Primavera P6 is a widely used construction project scheduling and controls platform that allows teams to create and maintain detailed schedules containing activities, milestones, durations, dependencies, critical paths, resources, and progress. As a project progresses, teams update actual dates, remaining durations, and other schedule information. P6 supports baselines and schedule versions that allow teams to compare current performance with earlier plans and, when auditing is configured, can record detailed changes to schedule fields. This historical schedule data can then be analyzed by tools such as SmartPM to identify the underlying drivers of project delays.

- **Root cause analysis problem:** SmartPM focuses on determining ***why construction projects and milestones are falling behind schedule and identifying the activities***, logic changes, execution problems, or other events responsible for schedule slippage. Its analytics help teams distinguish symptoms from the underlying causes driving changes in project completion dates, supporting schedule recovery, cost control, and delay or claims analysis.
- **Data processed:** SmartPM primarily analyzes structured CPM schedule data, including project baselines, successive schedule updates, activity dates, durations, dependencies, critical paths, constraints, milestones, float, and schedule changes. It supports schedule information from systems such as Primavera P6 and Microsoft Project and integrates with construction platforms including Procore and Autodesk.
- **Client coverage:** SmartPM is purpose built for construction and serves general contractors, owners, developers, and project controls teams. Publicly referenced customers include JE Dunn, Ferrovial, Manhattan Construction, Barton Malow, Suffolk, and AECOM. Its industry analysis has covered more than 120,000 CPM schedules.

4.3 Sologic Causelink

- **Root cause analysis problem:** Causelink addresses the challenge of conducting complex, evidence-based investigations consistently across teams, projects, and locations. It enables organizations to determine what happened, map causal relationships, validate potential

causes against available evidence, define corrective actions, and preserve investigation knowledge. Relevant use cases include construction safety incidents, quality failures, equipment problems, operational disruptions, and recurring systemic issues.

- **Data processed:** Causelink primarily processes human curated investigation information, including event timelines, observations, interviews, notes, photographs, videos, documents, and other supporting evidence. It organizes this information using cause and effect charts, 5 Whys, Ishikawa diagrams, and FMEA methodologies. Completed investigations can also be searched and compared to identify recurring causes and broader organizational patterns.
- **Client coverage:** Sologic serves organizations across construction, engineering, manufacturing, energy, healthcare, and other industries. A notable construction and engineering customer is Quanta Services, which has used Causelink Enterprise across decentralized operating units to standardize RCA practices and identify common causes across the organization.

4.4 Acerta.ai

- **Root cause analysis problem:** Acerta.ai focuses on identifying the manufacturing conditions responsible for product defects, test failures, process anomalies, scrap, rework, and reduced first time through yield. Complex manufacturing processes can contain thousands of interacting variables across machines and production stages, making manual RCA difficult. Acerta.ai applies machine learning to identify and rank the process signals most strongly associated with abnormal outcomes and trace quality problems back to upstream production conditions.
- **Data processed:** Acerta.ai processes high volume manufacturing and quality data, including machine and sensor signals, process parameters, tolerances, test results, product attributes, failure codes, and serialized part genealogy. Data can originate from PLCs, MES environments, test systems, and other production sources, allowing relationships to be analyzed across stations, production lines, and facilities.
- **Client coverage:** Acerta.ai is concentrated in complex discrete manufacturing, particularly automotive OEMs and Tier 1 suppliers. Publicly referenced customers and users include Ford, BMW, Dana, Linamar, ZF, BorgWarner, and Ballard, with applications covering components such as engines, axles, powertrain systems, and electric vehicle batteries.

4.5 Planera

Link: [Planera: How to Conduct Delay Analysis for Construction Projects: Methods, Claims and Best Practices](#)

- **Schedule risk and root cause analysis problem:** Planera focuses on identifying the activities, dependencies, constraints, and resource conditions responsible for construction schedule delays and project risk. Large construction schedules can contain thousands of interconnected activities, making it difficult for project teams to determine which upstream events are driving changes to critical paths, completion dates, or downstream work. Planera combines Critical Path Method (CPM) scheduling with AI-powered analysis to surface schedule risks, identify activities driving delays, trace schedule impacts through predecessor and successor relationships, and run “what-if” scenarios to assess potential corrective actions. Its AI scheduling assistant, Manny, allows users to interrogate project schedules in natural language, generate analyses and reports, and identify potential schedule issues.
- **Data processed:** Planera processes construction planning and project execution data, including activities, durations, milestones, predecessor and successor relationships, calendars, constraints, planned and actual dates, remaining durations, critical and float paths, labor and equipment resources, baselines, progress updates, variance reason codes, and schedule versions. The platform can analyze this information through its CPM engine, schedule quality checks, dashboards, Monte Carlo risk simulation, and AI assistant, allowing project teams to evaluate schedule health, understand changes over time, identify resource or schedule bottlenecks, and assess the potential impact of delays across a project.
- **Client coverage:** Planera is concentrated in the construction industry, serving general contractors and project teams across preconstruction, project execution, pursuits, and enterprise portfolio management. The company states that its platform is used by more than 100 builders. Publicly referenced customers and users include Balfour Beatty, Barton Malow, Granite Construction, Hensel Phelps, Ryan Companies, Brasfield & Gorrie, Garney Construction, Big-D Construction, Stacy Witbeck, Harkins, and Swinerton. Applications span project scheduling, field-to-master schedule coordination, schedule quality and compliance, resource planning, progress tracking, risk analysis, and portfolio-level schedule visibility.

Section 5. Schedule delays in construction

5.1 Causes for delay

Link: [Planera: How to Conduct Delay Analysis for Construction Projects: Methods, Claims and Best Practices](#)

Rank	Delay Driver	Reported Prevalence	Typical Schedule Impact
1	Planning & Scheduling Issues*	76%	Baseline slippage driven by unrealistic durations, dependencies, and constraints
2	Supply Chain & Material Procurement	65%	Long-lead materials and equipment delay installation and downstream activities
3	Labor Shortages	45%	Insufficient skilled labor reduces productivity and delays planned work
4	Design & Scope Changes	~45%	Late design decisions, RFIs, and scope changes create stoppages and rework
5	Permitting & Regulatory Approvals	~29%	Delayed permits, inspections, and approvals prevent work from proceeding
6	Weather & Site Conditions	~3–5%	Adverse weather and unforeseen conditions interrupt field activities

5.2 Critical Path: What Determines the Project Finish Date?

Critical Path is the longest sequence of dependent activities that determines the minimum time required to complete the project. Activities on this path have zero slack, meaning a delay to any critical activity directly delays the project completion date. Slack or Float is the

amount of time an activity can be delayed without delaying the overall project completion date. Activities with positive slack have some scheduling flexibility; activities with zero slack are critical.

5.3 Supplementary data: Superintendent daily logs and RFIs (Request for information)

Enterprise project portfolio management tools like Oracle P6 capture what was planned, when activities are scheduled, their logic, progress, and forecast dates; superintendent logs and RFIs provide the field context explaining *why* work progressed, slowed, stopped, or changed.

Information captured	Superintendent Logs / RFIs can reveal	What P6 schedule data typically shows
Weather & Site Conditions	Rain, snow, extreme temperatures, wind, flooding, ground conditions, and specific hours or activities affected	Activity dates and schedule variance, but generally not the detailed field conditions causing lost productivity
Safety Events	Incidents, near misses, stop-work events, safety meetings, hazards, corrective actions, restricted areas	Resulting schedule impact may appear, but not the underlying safety event or circumstances
Labor / Manpower	Daily crew counts by contractor/trade, hours worked, absenteeism, staffing shortages, and where crews were deployed	Planned resources if resource-loaded, but often lacks detailed actual daily field deployment
Equipment	Equipment on site, breakdowns, idle equipment, unavailable equipment, and operating hours	Scheduled activity/resource information, but generally not the detailed reason equipment was unavailable or idle
RFIs & Design Questions	What information was requested, when it was requested, when the response arrived, and which work was waiting for an answer	May show an impacted activity or milestone, but not necessarily the question, response, or decision history
Unforeseen Conditions	Unexpected utilities, unsuitable soils, obstructions, existing-condition conflicts, access restrictions	Schedule impact can be incorporated later, but the original field condition and discovery context may be absent

Material & Delivery Issues	Late/rejected deliveries, shortages, incorrect materials, delivery dates, and materials actually received	Procurement activities may be scheduled, but the detailed reason for a delivery disruption may not be captured
Inspections & Quality	Failed inspections, rework, testing results, deficiencies, punch-list issues, and waiting for inspectors	May show inspection activities or resulting delays, but often lacks the underlying quality/inspection detail
Work Stoppages & Disruptions	Exact start/stop times, affected crews, locations, cause of disruption, and work that could/couldn't proceed	Shows activity progress and resulting dates, but may not explain what happened during the day
Field Direction / Changes	Verbal directions, owner/architect instructions, changed work, sequencing changes, and pending change orders	Revised logic/durations may eventually reflect the change, but not necessarily who directed it or why
Actual Work Conditions	Work performed by location, productivity issues, trade stacking, congestion, access limitations, and out-of-sequence work	Shows activity-level progress, but not necessarily how the work was actually executed in the field
Photos & Observations	Timestamped visual evidence of progress, conditions, obstructions, deficiencies, and disruption	No equivalent visual record within the CPM schedule itself

REPORT NO:		DATE:	
CONTRACTOR:		PROJECT NO:	
PROJECT NAME:			
FOREMAN/SUPERVISOR:			
WEATHER:		SITE CONDITION:	
☐ Clear ☐ Windy ☐ Cool ☐ Overcast ☐ Fog ☐ Warm ☐ Rain ☐ Cold ☐ Hot		☐ Clear ☐ Dusty ☐ Muddy Temperature Range:	
DAY:		☐ Monday ☐ Tuesday ☐ Wednesday ☐ Thursday ☐ Friday ☐ Saturday ☐ Sunday	
SCHEDULE		PROBLEMS/DELAYS	
CONTRACT DATE:			
COMPLETION DATE:			
PROBABLE COMPLETION DATE:			
ACTUAL WORK COMPLETED %:			
NUMBER OF WORKERS:			
ACCIDENTS/SAFETY ISSUES:			
SUMMARY OF WORK / JOBSITE NOTES			
CONSTRUCTION SUPERINTENDENT DAILY LOG BOOK			
SIGNATURE		NAME	

Fig. 6 Example Superintendent daily log book

5.4 Schedule delay analysis methods

Link: [Planera: How to Conduct Delay Analysis for Construction Projects: Methods, Claims and Best Practices](#)

- **As-Planned vs. As-Built:** Compares the **original baseline schedule with actual project performance** to identify overall delay. Simple and cost-effective, but limited in determining the specific causes of delay.
- **Impacted As-Planned:** Adds **delay events to the original baseline** to estimate their potential impact on project completion. Easy to model, but does not reflect actual progress or subsequent schedule changes.
- **Time Impact Analysis (TIA):** Inserts specific **delay events (fragnets) into the contemporaneous CPM schedule** to measure their impact on completion. Strong for event-specific analysis but requires reliable schedule updates and detailed records.
- **Windows Analysis:** Breaks the project into **defined time periods** to evaluate how delays and the critical path changed over time. Well suited to complex projects but requires significant data and analysis.
- **Collapsed As-Built (But-For):** Starts with the **completed as-built schedule and removes selected delay events** to estimate when the project would have finished without them. Useful for post-completion analysis but highly dependent on reliable as-built records.

Section 6. Causal inference and RCA

Link: [Causal Inference and Root Cause Analysis in Complex Systems](#)

- **RCA vs. causal inference:** **Causal inference** is the process of determining the independent effect of a cause on an outcome by asking how the outcome would change if the cause were changed, rather than just observing their correlation. **Root Cause Analysis (RCA) is broader and more action oriented: it asks why a specific failure occurred and what intervention can prevent it from recurring.** In complex systems, there may be multiple interacting causes rather than a single root cause, so identifying a causal relationship alone may not identify the best corrective action.
- **Using causal inference for RCA:** Causal inference can strengthen RCA by testing which suspected causes actually influence the failure outcome. Teams can first develop hypotheses or a causal graph linking potential causes to the failure, then use statistical methods, controlled experiments, natural experiments, or counterfactual analysis to estimate what would happen if a suspected cause were changed. For example, rather than concluding that high temperature causes manufacturing defects because the two are correlated, causal analysis can test whether changing temperature would actually reduce defects while controlling for other process variables. Causal inference therefore provides evidence for causal links, while RCA determines which interventions should be implemented.

- **Data needed for causal inference:** Causal inference generally requires historical observational or experimental data containing the failure outcome, suspected causal variables, and relevant confounding variables. For manufacturing RCA, this could include machine and sensor readings, process parameters, material characteristics, timestamps, product genealogy, operator or shift information, environmental conditions, maintenance events, and quality or defect outcomes. Ideally, the dataset contains sufficient variation in operating conditions and examples of both successful and failed outcomes. Domain knowledge is also important for constructing the causal graph and deciding which variables may be causes, confounders, mediators, or consequences.

Section 7. Example case study for root cause analysis: Building collapsed due to structural failure

Link: [How to use Root Cause Analysis \(#RCA\) in the #construction sector?](#)

Problem: A building under construction collapsed due to structural failure.

Step 1: Define the problem.

The building was a 12-story residential complex. It collapsed on Saturday night at 11 pm. It was under construction for six months and was scheduled to be completed in three months. The collapse killed four workers and injured six others. It also damaged nearby properties and vehicles.

Step 2: Collect data.

The data and evidence collected included:

- The design and construction documents of the building
- The inspection and quality control records of the building
- The soil and foundation reports of the site
- The weather conditions and seismic activity at the time of the collapse
- The eyewitness accounts and testimonies of the workers, residents, and authorities
- The forensic analysis and investigation reports of the collapse
- The debris and wreckage of the collapsed building

Step 3: Identify possible causes.

- The possible causes brainstormed using a fault tree analysis were:

Step 4: Analyze possible causes.

- The possible causes were tested against the data and evidence collected using a Pareto chart. The results were:

Step 5: Identify root cause(s)

The root cause(s) identified were:

- The design of the building was faulty and did not comply with the codes and standards.
- The construction of the building was poor and did not follow the specifications and best practices.
- The supervision and quality control of the building was inadequate and did not detect or correct the errors and defects.

Step 6: Recommend solutions.

The solutions recommended were:

- Reviewing and revising the design of the building and ensuring its compliance with the codes and standards
- Demolishing and rebuilding the building according to the revised design and specifications
- Hiring more qualified and experienced engineers, contractors, and inspectors
- Implementing more rigorous supervision and quality control procedures and standards
- Providing more training and awareness programs for workers and stakeholders

Step 7: Implement solutions.

The solutions implemented were:

- Hiring an independent consultant to review and revise the design of the building
- Obtaining necessary permits and approvals from authorities and stakeholders
- Following safety protocols and best practices during demolition and reconstruction activities
- Documenting and communicating the progress and results of the solutions
- Evaluating and measuring the performance and reliability of the building after completion

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- [Five principles of RCA](#)

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- [FM-Water Damage and Escaped Liquids](#)

- [Zurich-Water Damage Mitigation Brochure PDF](#)
- [Zurich-Water Damage Mitigation Fact Sheet PDF](#)
- [Chubb-Preventing Water Damage](#)
- [Chubb-Smart Water-Damage Prevention](#)
- [Nationwide-Construction Water Mitigation Program](#)

7. Limits of RCA method

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